

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP30019
Subject Title	Data Analysis Techniques for Scientists
Credit Value	3
Level	3
Pre-requisite / Co-requisite/ Exclusion	COMP1012/ AP20005
Objectives	To introduce basic techniques in numerical computation, data analysis, data modeling, and their applications to practical problems in science and engineering.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) formulate problems in science and engineering in mathematical forms; (b) apply numerical methods for solving mathematical problems relevant to science and engineering; (c) demonstrate proficiency with statistical analysis of data; (d) formulate and apply appropriate models of data analysis to solve practical problems.
Subject Synopsis/ Indicative Syllabus	Solution of nonlinear equations: fixed-point iteration method, bisection method, Newton-Raphson method. Approximation of data: linear interpolation, Lagrange interpolation, Taylor series, approximation of derivatives using finite difference schemes. Random numbers: probability, probability density function, discrete and continuous random variables, uniform and non-uniform pseudo random numbers, exponential variables, Gaussian variables. Statistical inference: mean, standard deviation, variance and covariance, sum of random variables, law of large numbers, confidence level, hypothesis testing. Regression: linear regression, gradient descent, momentum, stochastic gradient descent, nonlinear regression, logistic regression.
Teaching/Learning Methodology	Lecture: To introduce basic techniques in numerical calculations, data analysis, and data modeling. Demonstrations on solution of practical problems will be conducted. Students are encouraged to raise questions when meeting difficulties. Computer laboratory: Students work on given problem sets either individually or through interaction among each other. They are encouraged to raise questions and discuss any issues with the instructor. These problem sets provide the opportunities to apply the knowledge gained from the lectures and to consolidate what have been learned.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	(1) Continuous assessment	40	✓	✓	✓	✓
	(2) Examination	60	✓	✓	✓	✓
	Total	100				
	The continuous assessment is based on computer laboratories, assignments and a mid-term test. The examination is a three-hour written final examination. Various kinds of questions will be set in both components to assess the intended learning outcomes.					
Student Study Effort Expected	Class contact:					
	• Lecture			26 h		
	• Laboratory			18 h		
	Other student study effort:					
	• Self-study			76 h		
	Total student study effort			120 h		
Reading List and References	Textbook:					
	L. Igual, S. Segu, “Introduction to Data Science. A Python Approach to Concepts, Techniques and Applications”, Springer (2017).					
	References:					
	C. Shah, “A Hands-On Introduction to Data Science”, Cambridge University Press (2020)					
	C.M. Bishop, “Pattern Recognition and Machine Learning”, Springer (2006).					
	J Zelle, “Python Programming: an Introduction to Computer Science”, 3rd Edition, Franklin, Beedle & Associates (2016).					
	A. Scopatz and K.D. Huff, “Effective Computation in Physics: Field Guide to Research with Python” O’Reilly Media (2015).					